Demo Client · DEMO-0004
11 November 2026 · v1.0

Field	Value
Document	Methodology for the development, validation and monitoring of a retail credit scorecard
Template version	credit_scoring 1.0.0 / proquant-core 0.1.0
Engagement	DEMO-0004 · Demo Client
Model	Retail Personal Loan Scorecard, Unsecured personal loans (synthetic example)
Prepared by	ProQuant Analytics · info@proquantanalytics.com
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1. Purpose and scope

The scorecard ranks applicants by the likelihood of default so that the lender can approve, refer or decline, and estimates a probability of default (PD) for each. This document describes how the model was built, validated, turned into a decision strategy, tested for fairness and is to be monitored. It covers application scoring of a retail portfolio; it does not cover behavioural scoring, collections, pricing models or regulatory capital models (IRB), which have their own requirements.

2. Data

2.1 Inputs

Data set	Size	Fingerprint (SHA-256)
applications	68,253 rows	`4066bc75c1d0ce2c`
performance	46,366 rows	`b728ddc6477b9f52`

Each input is fingerprinted with SHA-256 on ingest and recorded in the run audit trail.

2.2 Outcome definition and maturity

Bad = 90+ days past due within the outcome window (default 12 months) of booking. An outcome is mature only when the whole window has elapsed at the performance cut-off date. Immature loans are excluded from model building and validation (they would understate the bad rate) and are scored only for monitoring.

2.3 Cleaning rules (all logged in `tables/cleaning_log.csv`)

1. Rows without an application id or date are dropped; exact duplicates are dropped; for repeated ids the latest row is kept.
2. Category spellings are harmonised ignoring case and spaces (most frequent spelling kept).
3. Values outside the business range of a field become missing (never a guess).
4. Outcome rows without a matching application, or with an invalid flag, are dropped.
5. Booked loans past the outcome window without an outcome are excluded and reported.

6. Declined applications have no outcome: they are counted (approval rate) and not modelled.

2.3 Samples

Assigned by application date: **development** (split at random, stratified by outcome, into training and an in-time test), **out-of-time validation** (a later period never used for fitting or tuning) and **monitoring** (later still). Bin boundaries, coefficients and the cut-off are derived from the development training sample only (the cut-off from the out-of-time sample's outcomes, see section 7).

3. Characteristics: weight of evidence and information value

Numeric characteristics start from quantile bins; bins smaller than the minimum share are merged with the neighbour of the closest bad rate; where a monotonic bad-rate trend is expected (direction taken from the data unless configured) adjacent bins that violate it are pooled; the number of bins is capped. Categories are grouped (rare ones pooled in "Other", which also receives categories unseen in development). Missing values have their own bin; when fewer than 30 training rows are missing the bin gets a neutral WoE of 0.

```
WoE(bin) = ln( (goods in bin + 0.5) / total goods ) − ln( (bads in bin + 0.5) / total bads )
IV(characteristic) = sum over bins of (share of goods − share of bads) x WoE
```

Rule of thumb: IV below 0.02 useless, 0.02 to 0.1 weak, 0.1 to 0.3 medium, 0.3 to 0.5 strong, above 0.5 suspicious (check for leakage: information not available at the time of the decision).

4. The scorecard

Logistic regression of the bad flag on the WoE-coded characteristics:

```
ln( P(bad) / P(good) ) = b0 + sum_i b_i x WoE_i
```

Selection: IV at least `selection.min_iv`; of two characteristics correlated above `selection.max_corr` the one with higher IV is kept; then backward elimination removes characteristics with a coefficient of the wrong sign (positive: more WoE must mean less risk) or a p-value above `selection.p_value`. Multicollinearity is reported as the variance inflation factor (VIF).

Scaling: `factor = pdo / ln 2`, `offset = base_score − factor x ln(base_odds)`, score = offset + factor x ln(odds good:bad). The points of bin j of characteristic i are `−(b_i x WoE_ij + b0 / n) x factor + offset / n`, rounded to integers, so the score is the sum of the bin points. PD = 1 / (1 + odds good:bad). Scores are limited to the configured range.

Reason codes: for each applicant, the characteristics ranked by points lost relative to the best bin (adverse-action reasons).

5. Validation

Metric	Definition
AUC	probability that a random bad has a higher risk than a random good
Gini	2 x AUC - 1
KS	maximum distance between the cumulative distributions of goods and bads
Lift	bad rate of a score decile divided by the portfolio bad rate
Calibration	observed bad rate versus mean predicted PD by PD band; ratio observed / predicted; binomial p-value per band; Hosmer-Lemeshow

Gini and KS are reported on training, in-time test and out-of-time samples, with a bootstrap 95% interval (200 resamples). Tests (PASS / WARN / FAIL against thresholds in the configuration): discrimination, stability (training to out-of-time), overfitting (training to test), coefficient signs, VIF, leakage screen (largest IV), sample size, calibration, drift, challenger gap, fairness, data quality. Any FAIL gives "Not approved for use"; any WARN "Approved for use with conditions". Drift tests are monitoring signals: a breach is a WARN asking for action and does not by itself withdraw the approval.

Challenger. A gradient-boosting model (histogram-based, shallow trees, early stopping) on the same characteristics, never on protected attributes. It measures what the transparent scorecard gives up; permutation importance on the out-of-time sample ranks its characteristics.

6. Risk grades

Five grades A (safest) to E at the 20th, 40th, 60th and 80th percentiles of the development scores, with the bad rate of each grade measured out-of-time.

7. Cut-off strategy

For each candidate cut-off, applicants at or above it are approved. Using the observed out-of-time outcomes:

`good loan: + amount x net margin x term / 24` `bad loan: − amount x LGD`

(net margin is annual after funding and operating costs; the average outstanding balance is taken as half the amount). The recommended cut-off maximises total profit subject to the bad-rate appetite of the approved book. Sensitivity shows the cut-off under LGD +-10 points and margin +-3 points. A refer zone of `strategy.refer_width` points below the cut-off is sent to manual review. **Limitation:** outcomes exist only for granted loans (no reject inference), so cut-offs below the historical acceptance region are extrapolations.

8. Fairness

At the cut-off, for each protected attribute and group: approval rate; adverse impact ratio (AIR) = approval rate / approval rate of the best-treated group (below 0.80 is the "four-fifths" flag); approval rate among goods (equal opportunity) and bads; observed / predicted bad rate; AUC. Status PASS at AIR >= `air_pass`, WARN at >= `air_warn`, otherwise FAIL; groups smaller than `min_group_size` are reported, not rated. Proxy analysis measures the association (Cramer's V) between each characteristic and each protected attribute. When a group is flagged, a less discriminatory alternative is searched: the scorecard is re-fitted without the characteristic most associated with the attribute, and Gini, approval rate and AIR are compared at the same overall approval rate. Protected attributes are never model inputs. A gap does not prove discrimination, and clearing a test is not legal advice; the applicable standards depend on the jurisdiction and the product.

9. Monitoring

Quarterly, against the development training sample:

`PSI = sum over bins (share_now − share_dev) x ln(share_now / share_dev)`

on the score (development deciles) and CSI on each scorecard characteristic (the scorecard's own bins). Below 0.10 stable, 0.10 to 0.25 watch, above 0.25 breach. Also: categories not seen in development, and Gini, bad rate and calibration by quarter on mature outcomes (which lag by the outcome window). Persistent breaches lead to recalibration (intercept shift) or redevelopment.

10. Assumptions, limitations and deviations

Granted loans only; fixed bad definition and outcome window; lender-supplied economics; fairness on available attributes only; significance tests reject small deviations in large samples. No deviations from the standard methodology.

11. Reproducibility

Each run folder contains `config_used.yaml`, `audit.json` (analyst, timestamps, code version, input fingerprints, output hashes), `run.log`, every result table as CSV and all documents. Random seeds are fixed (sample split, bootstrap, challenger), so re-running with the same inputs reproduces every figure. The scoring pack's Python scorer is tested against the pipeline's scores at build time.

12. References

- Siddiqi, N. (2017). *Intelligent Credit Scoring*, 2nd ed. Wiley.
- Thomas, L., Edelman, D., & Crook, J. (2002). *Credit Scoring and Its Applications*. SIAM.
- Anderson, R. (2007). *The Credit Scoring Toolkit*. Oxford University Press.
- Basel Committee on Banking Supervision (2005). *Studies on the Validation of Internal Rating Systems*, Working Paper 14.
- Hosmer, D., Lemeshow, S., & Sturdivant, R. (2013). *Applied Logistic Regression*, 3rd ed. Wiley.
- Hand, D. (2009). Measuring classifier performance: a coherent alternative to the AUC. *Machine Learning*, 77.
- Equal Credit Opportunity Act, Regulation B (adverse action and disparate impact); EU Consumer Credit Directive 2023/2225.